Customer Stories / Agriculture

2024

Sencrop Improves Sustainability in Farming by Using AWS for Its In-Field Data Application

Using AWS, Sencrop built a microclimate application that uses ML to provide more accurate weather and climate data to farmers to help them implement sustainable agricultural practices.

Overview | Opportunity | Solution | Outcome | AWS Services Used

1.5 billion

weather data points gathered daily

80 TB

68%

of farmers saved farm vehicle trips

Overview

Farmers rely on accurate weather predictions to optimize crop yields and implement sustainable agricultural practices. The problem is that accurate forecasts aren't always so easy to come by; even rainfall predictions are right only 40 percent of the time. Sencrop, a smart agriculture startup based in France, wanted to create a solution that would enhance the accuracy of weather predictions and give farmers access to precise climate condition insights to support data-driven decision-making.

Sencrop built its microclimate application on Amazon Web Services (AWS) using core services such as Amazon EMR—a cloud big data solution for petabyte-scale data processing, interactive analytics, and machine learning (ML). Using Sencrop's solution, almost 30,000 farmers across Europe improved sustainability efforts by reducing water use, chemical spray treatment frequency, and tractor and harvester trips.

Opportunity | Building a Data-Driven Microclimate Application on AWS

When building its app, Sencrop needed to find a way to gather and analyze large amounts of data from radar, weather forecasts, and nearly 40,000 in-field sensors, and then present insights to farmers in a readable, easy-to-interpret way. It also wanted its app to integrate input from farmers directly, allowing them to share their observations with other users to foster a community of sustainability-minded growers.

Initially, Sencrop used AWS microservices to deploy infrastructure quickly and efficiently, but it has since expanded its use of AWS to take advantage of the broad range of services and low costs of operation and maintenance.

Solution | Helping Farmers Improve Sustainability Using AWS

Sencrop uses Amazon EMR to process data in near real time. The data is then stored using Amazon Aurora, a relational database management system built for the cloud. Sencrop applies ML to analyze the data as it is collected, using AWS for the compute power needed to handle the enormous amounts of data to train its ML models. Using AWS, Sencrop's app ingests 1.5 billion data points from weather forecasts daily and processes a total of 80 TB of data in its system. The company recently launched the Sencrop Forecast feature, which uses ML to analyze weather forecasts to create accurate, hyperlocalized forecasts for its users.

From a representative sample of over 1,000 farmers, 90 percent of vine and potato growers reported reduced chemical spray treatment frequency, 68 percent of farmers saved farm vehicle trips, 43 percent of farmers using Irricrop saved water, and 76 percent of respondents declared Sencrop helped them in their agro-environmental practices (for example, sustainable agriculture, soil conservation, or regenerative farming).

Outcome | Expanding into a Global Farming Solution Using AWS

To further improve its data analysis to benefit farmers, Sencrop is conducting research and development projects in France using AWS and ML. The company is also planning to expand its solution to new regions.

"Using AWS, we can expand to a new region in a new part of the world and keep our architecture exactly the same," says Mathieu Despriée, chief technology officer of Sencrop. "We know that there will be good connectivity and that everything will be congruent."

About Sencrop

Sencrop provides a microclimate application to help farmers produce higher crop yields and develop sustainable practices. Its app allows farmers to access precise climate and observational data so that they can optimize decision-making in the field.

AWS Services Used

Amazon Aurora

Amazon Aurora provides built-in security, continuous backups, serverless compute, up to 15 read replicas, automated multi-Region replication, and integrations with other AWS services.

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Amazon EMR

Amazon EMR is the industry-leading cloud big data solution for petabyte-scale data processing, interactive analytics, and machine learning using open-source frameworks such as Apache Spark, Apache Hive, and Presto.

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2024

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